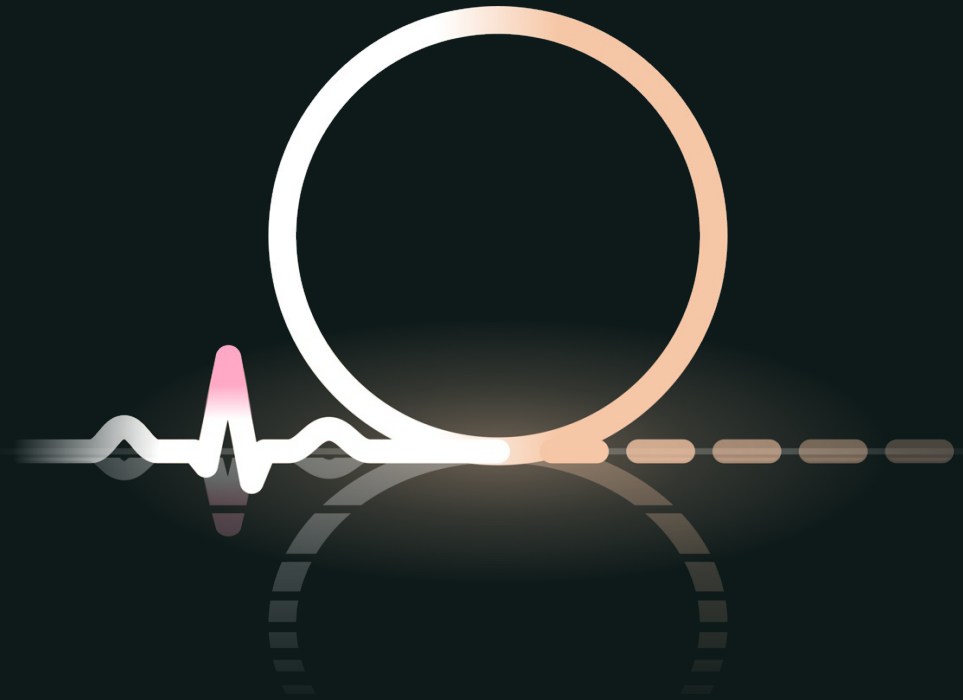
Pratibimb

प्रतिबिम्ब means reflection

An early-warning digital twin for hypertension, built on the watches India already owns

Fuses the health record with a budget watch and a home BP cuff, learns how one body responds to sleep and walking, and warns the doctor up to 72 hours before a cardiovascular strain episode.



Team AMAZE

Nikita Pandey (team leader), Suraj Aryan

IIM Lucknow, PGPSM | Happiest Health Digital Twin Challenge 2026

Live demo **pratibimb-pi.vercel.app**

Code github.com/pndynkt/Team-AMAZE_IIM-Lucknow

Pratibimb at a glance

Watch, home BP cuff and health record, read through a personal twin and scored every night

Problem

- **1 in 4 adults** in India live with high blood pressure
- **Strain builds for days** before a surge, clinic or ER visit
- **Fixed thresholds** flag 55.8% of readings; doctors stop looking

Solution

- **Personal twin** learns how one body reacts to sleep and walking
- **Alerts only** on change her own habits cannot explain
- **Calibrated 72 h risk** with reasons in plain words

Impact

- **70% of episodes** caught at 1 false alert per 22 patient-days
- **Median 48 h** warning before the event
- **Twin lift** +0.025 AUROC, better on 3 of 3 splits

Implementation

- **Working product:** doctor console, patient app in English and Hindi
- **FHIR R4 and ABDM** WellnessRecord export
- **Live demo,** 25 tests, MIT licence

0.876
AUROC on patients the model never saw

70%
episodes caught inside the false-alert budget

48 h
median warning before the event

240 x 120
synthetic patients x days, seeded

Hypertension builds quietly; the watch sees it, nobody reads it

Four facts that shape the design

1 in 4

Indian adults

have high blood pressure. Most strain builds for days before a surge, clinic visit or ER trip.

91.6%

of smartwatches

sold in India cost under US$50 (IDC, Q2 2026). They log heart rate, sleep and steps every night.

55.8%

of readings flagged

by fixed thresholds in one remote-monitoring study. Alert fatigue follows.

2 vs 0.3

bpm per short night

One person's heart rate rises 2 bpm, another's 0.3. A population rule cannot tell them apart.

The gap: nobody turns cheap nightly watch data into a personal, calibrated warning a doctor can act on, ahead of time and inside a false-alert budget. **Pratibimb closes that gap.**

Predict a strain episode 72 hours ahead, then hand it to people

Healthcare use case: early warning for adults with hypertension, read by their own doctor

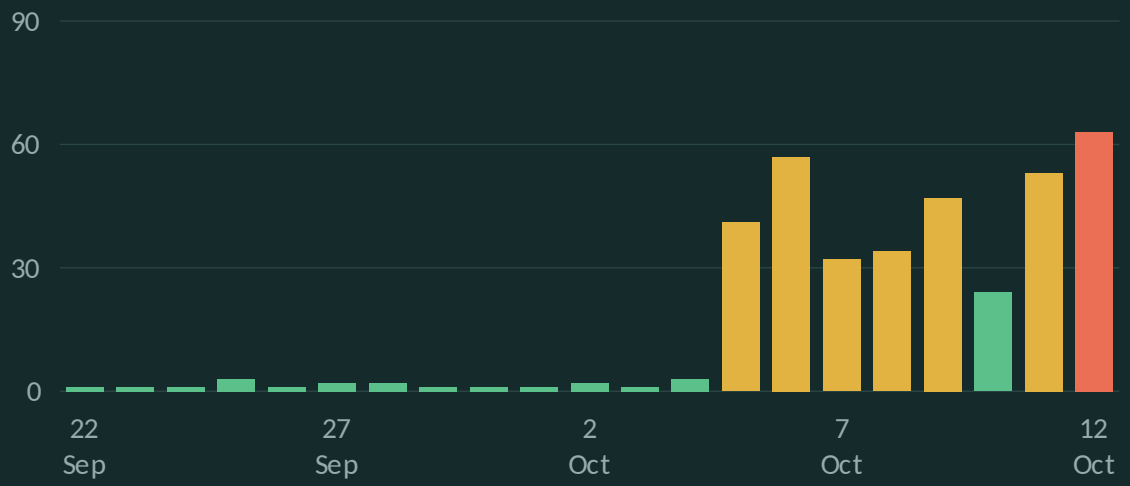
The event we predict

- **Overnight heart rate** above her own 28-day baseline
- **HRV falls** below her smallest worthwhile change
- **Sleep and steps** drop; home systolic trend rises over 7 days
- **Ends in an event day:** hypertensive surge, unplanned clinic or ER visit

Who acts on it

- **Doctor:** triage by 72 h risk, reasons, what-if, acknowledge or escalate
- **Patient:** one sentence in English or Hindi, the habit that helps most, check-in, SOS

Meera Iyer, 58, Bengaluru: nightly risk score, last 21 nights



Gridlines at 30 (watch) and 60 (urgent). 9 and 10 Oct: watch off, the cuff kept reporting and the score kept updating.

Amber tier: the model
days ahead, calibrated, inside a budget

Red tier: people and the phone
SOS, contacts, one tap to 112

Two streams, one person: static record fused with the nightly watch

Every field below exists in the synthetic cohort and in the FHIR export

Health record (static EHR)

- **Demographics:** age, sex, city, masked ABHA
- **Past diagnoses:** SNOMED CT (hypertension, IHD, MI, diabetes, CKD)
- **Lab results:** LOINC (HbA1c, LDL, HDL, eGFR, creatinine)
- **Medicines:** class, dose, adherence log
- **Genetic marker:** polygenic risk tier; family history

Fused in the twin

- **Record reads the watch:** beta blockers dampen heart rate; diabetes blunts it
- **Twin removes** what her own sleep and steps explain
- **Fusion model joins** twin signals, BP trend and 8 record facts

Watch and cuff (dynamic)

- **Heart rate:** overnight resting, 5-minute intraday trace
- **HRV:** RMSSD against a 60-night baseline
- **Sleep stages:** duration, efficiency, deep, REM
- **Step count,** SpO2, respiratory rate
- **Home BP cuff:** morning systolic and diastolic

240 synthetic patients

120 days each

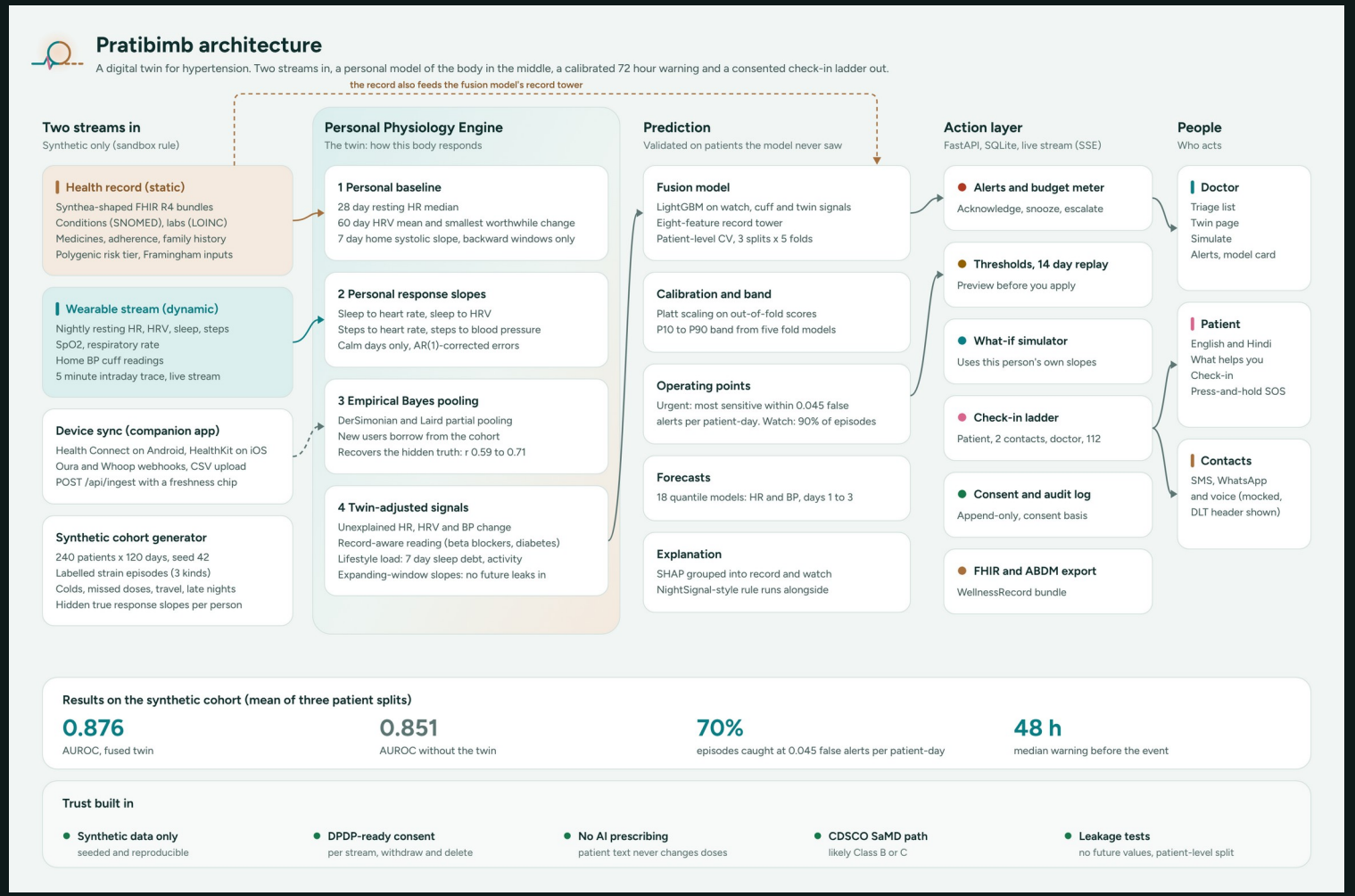
28,685 patient-days

342 labelled strain episodes

Planted confounders: 242 colds, 767 missed doses, 756 late nights, 290 travel days. **CGM:** not needed for hypertension; diabetes is read from the record (HbA1c).

Architecture: two streams in, a twin, a warning out

Full resolution in docs/architecture.pdf



How to read it

- **Two streams in:** FHIR record, watch and cuff via a companion app
- **Physiology engine:** baseline, slopes, pooling, twin-adjusted signals
- **Prediction:** LightGBM fusion, calibration, budgeted thresholds, forecasts, SHAP
- **Action layer:** FastAPI, SQLite, live stream; alerts, what-if, ladder, audit
- **People:** doctor console, patient app, consented contacts

How the twin decides: five steps, worked on Meera

Every number is computed by the trained model for Meera Iyer on 12 October

1

Personal baseline

Tonight vs her own last 28 nights, not vs other people

+13.8 bpm

above her usual 56.5 (tonight 70.3)

2

Response slopes

Calm-day regression of tomorrow's HR, HRV, BP on today's sleep and steps

4 slopes

per person, from 109 calm days

3

Empirical Bayes pooling

Her slope pulled toward the cohort by exactly its uncertainty

57%

weight on her own data; twin slope -1.41

4

Twin-adjusted signal

Tonight's change minus what her own habits explain

+12.6 bpm

unexplained; BP +16 mmHg unexplained

5

Fuse, calibrate, decide

LightGBM, Platt calibration, false-alert budget

24%

in 72 h (usual 0.7%); score 63, urgent

- **Calm days only:** episodes never teach the twin
- **AR(1)-widened errors:** honest error bars
- **Expanding window:** no future data; 3 leakage tests
- **Monotone model:** more sleep or walking never raises risk

Explained or not: a bad night versus a real warning

Steps 3 and 4 in detail, with Meera's numbers

Tonight's heart rate, split

Tonight vs her usual

+13.8 bpm

Explained by her habits

+1.3

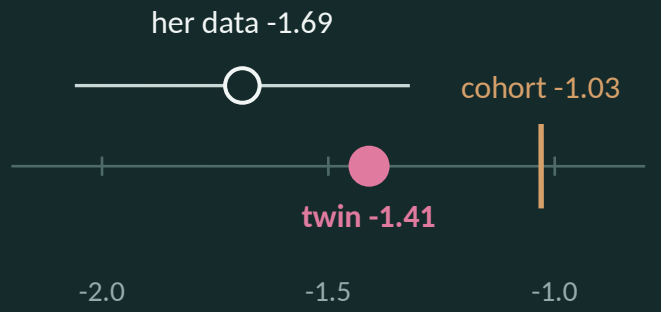
Unexplained: the warning

+12.6 bpm

- **Explained** = her slopes x her habits
- **Last night** 1.38 h less sleep, 1,498 more steps
- **(-1.38) x (-1.38) + (-0.43) x (1.50)** = +1.3 bpm
- **Home BP:** +16 mmHg her activity cannot explain

Borrowing strength from the cohort

Sleep to heart rate, bpm per extra hour of sleep



Weight on her own data
= spread² / (spread² + noise²)
= 0.43² / (0.43² + 0.37²) = 57%
57% x (-1.69) + 43% x (-1.03) = -1.41

New users start at the cohort value; the more calm days they log, the more the slope becomes their own.

Meera vs most people

Slope	Meera	Most
Sleep to HR	-1.41	-1.03
Walking to HR	-0.44	-0.32
Walking to BP	-1.28	-1.25
Sleep to HRV	+3.8%	+4.4%

Per extra hour of sleep or 1,000 steps; HR in bpm, BP in mmHg.

Her strongest lever: walking
30 min a day: BP -4.4 mmHg on her slope vs -4.3 on the cohort's; heart rate -1.5 bpm.

From signal to a calibrated, budgeted alert

Step 5 in detail: the fusion model, its guardrails and how it was validated

Inputs

- **Twin-adjusted** HR, HRV and BP
- **7-day** home BP trend
- **Load:** sleep debt, activity
- **8 record facts**

LightGBM

- **Gradient-boosted** trees
- **Monotone:** sleep and walking never raise risk
- **Prior MI, Framingham** only raise it

Calibrate

- **Platt scaling:** "20%" means about 20%
- **5 fold models** give the likely range

Budgeted tiers

- **Urgent:** most sensitive cut within 0.045 false alerts per patient-day
- **Watch:** 90% of episodes
- **Score 0 to 100:** 30 watch, 60 urgent

Explain

- **SHAP reasons** in plain words
- **Split** into record and watch
- **Suggested next step** for the doctor

Validation

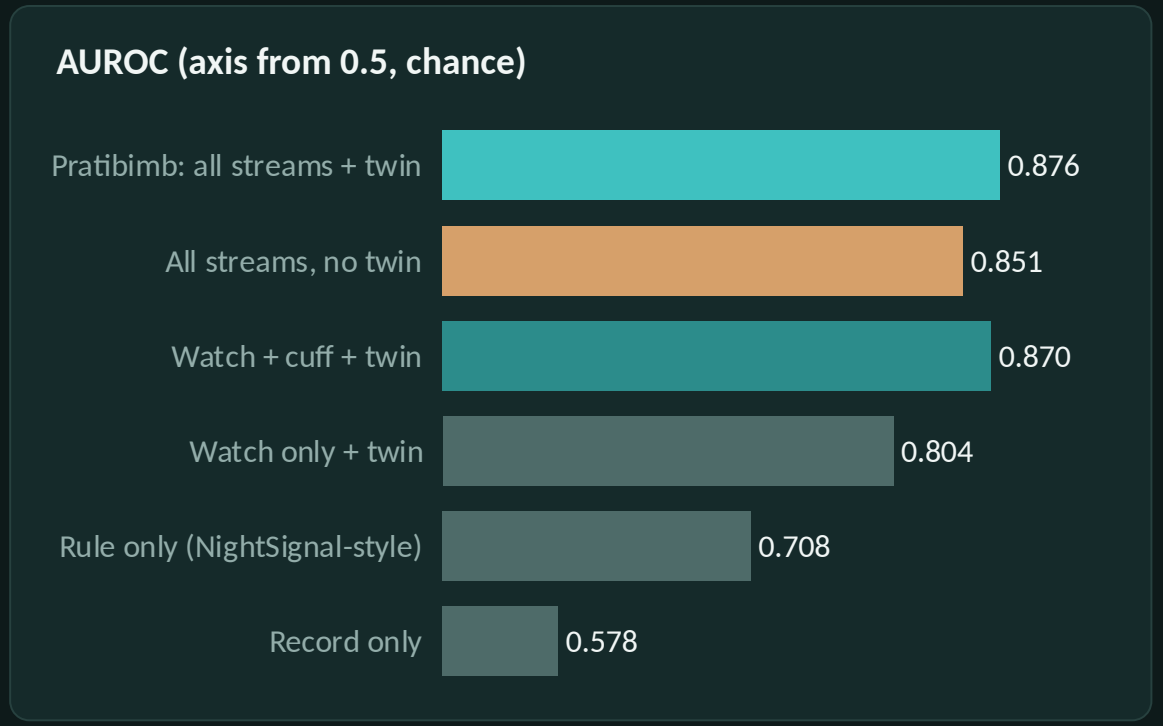
- **Patient-level 5-fold CV,** repeated on 3 splits (seeds 42, 43, 44)
- **Every dashboard patient** scored by a model that never saw them
- **3 leakage tests;** NightSignal-style rule run alongside as a baseline

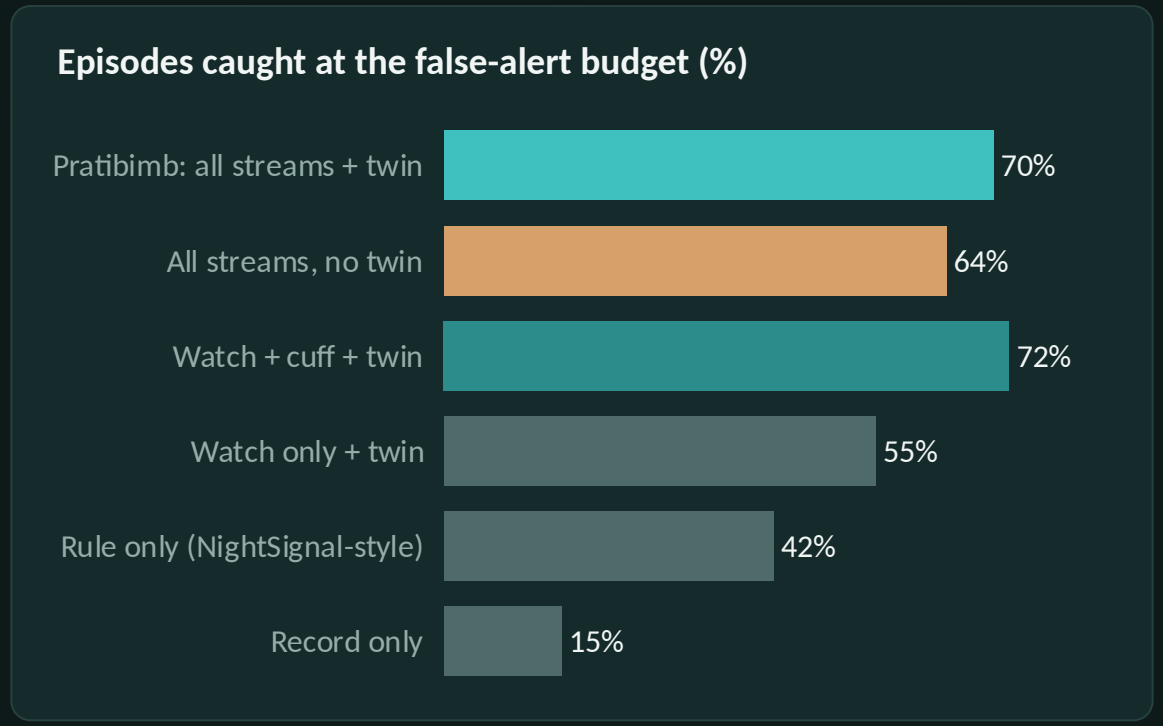
Forecasts for the next 3 days

- **18 quantile models:** resting HR and systolic BP, days 1 to 3, P10, P50, P90
- **Day-1 error:** 2.5 bpm heart rate, 6.2 mmHg BP
- **P10 to P90 band** holds 73% of next-day readings

Results: the twin adds lift on every patient split

Mean of three patient splits, 240 patients, measured on patients the model never saw





+0.025

AUROC from the twin, better on 3 of 3 splits; +7 points of episodes caught

48 h

median warning; half of all episodes caught 48 h or more ahead

1 per 22

patient-days: the false-alert budget (0.045 a day)

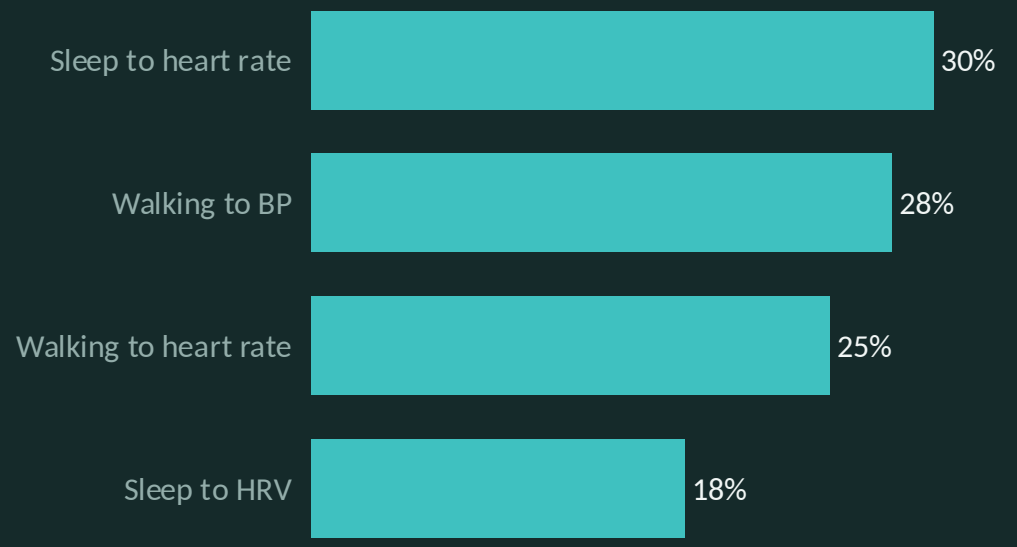
233 to 164

cold-like windows alerted at 90% sensitivity once the cuff joins

Proof the twin learns the right person

The generator hides each person's true slopes, so recovery can be checked directly

Error cut vs one slope for everyone (%)



Recovery of the hidden truth

Response	Match (r)	Twin error	One slope error	In 80% band
Sleep to heart rate	**0.71**	0.28	0.40	72%
Walking to heart rate	**0.66**	0.09	0.12	76%
Walking to BP	**0.67**	0.30	0.42	74%
Sleep to HRV	**0.59**	1.32	1.60	75%

Error: mean absolute error in each slope's own unit. One slope: the population average for everyone.

Personal what-if
Walking advice sized to her body, not the average body

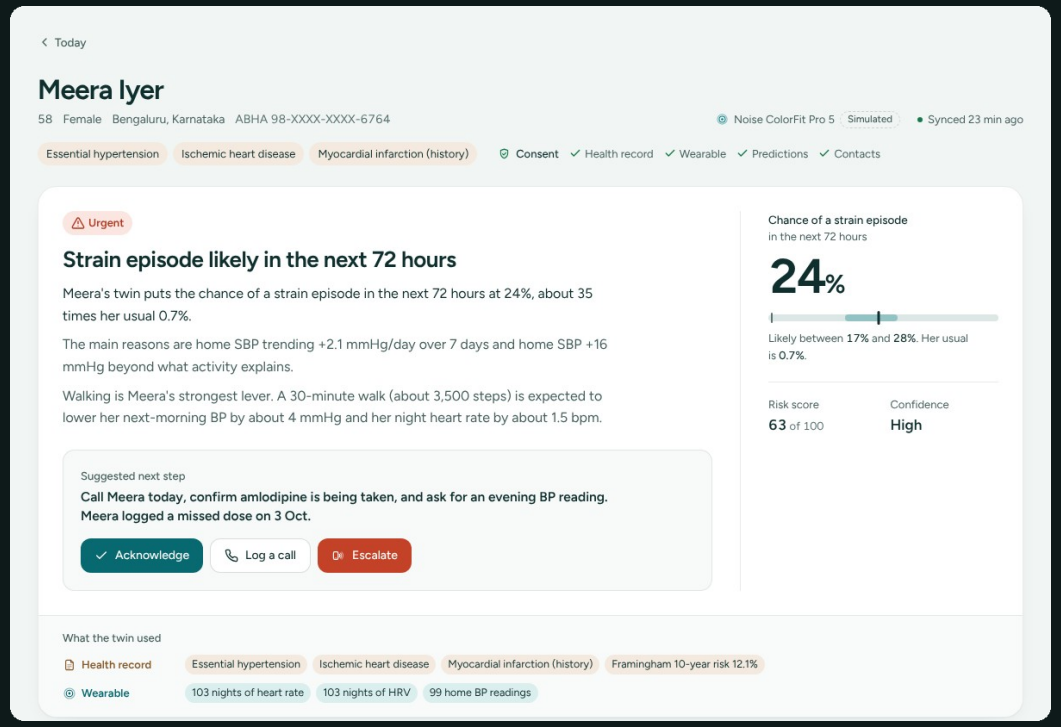
Fewer false alarms
A late night is explained by her own sleep slope, not alerted

Honest uncertainty
80% intervals hold 72% to 76% of the true slopes

Doctor console: who needs attention today, and why

Live demo: pratibimb-pi.vercel.app/console



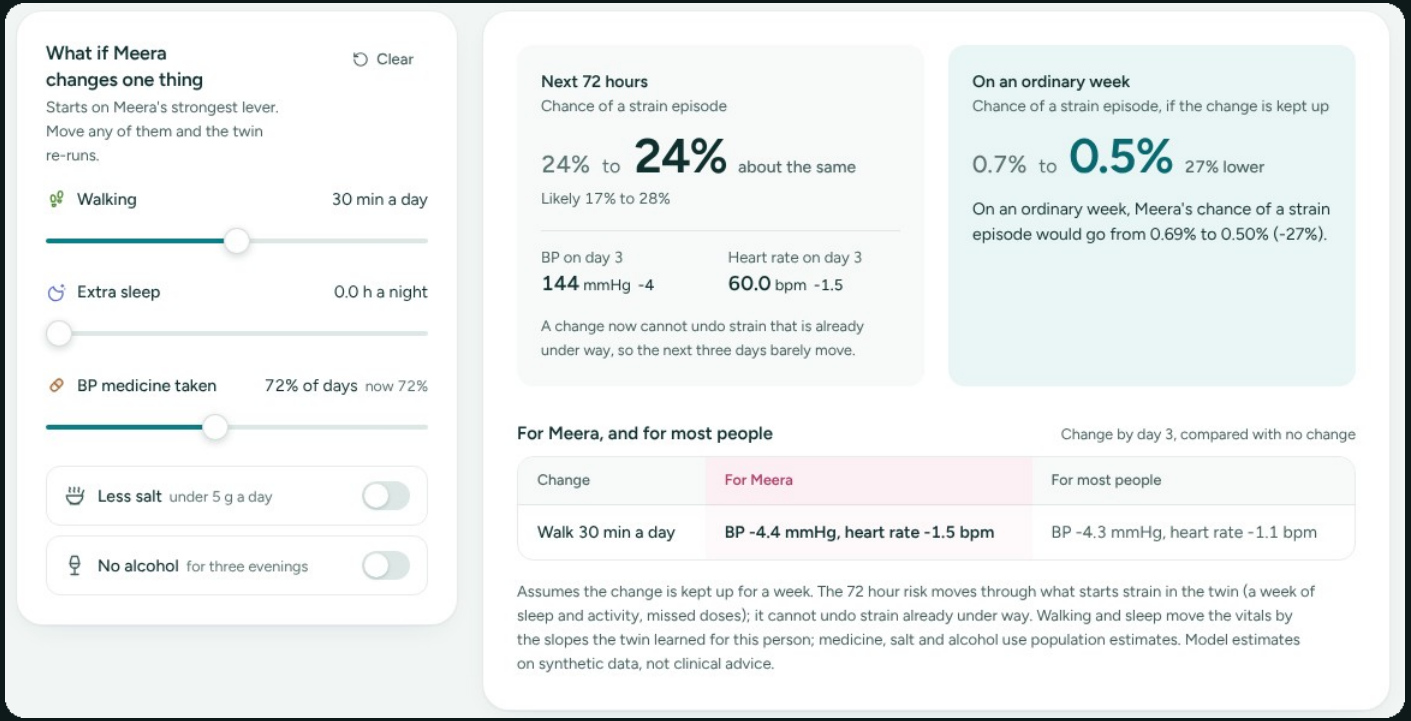


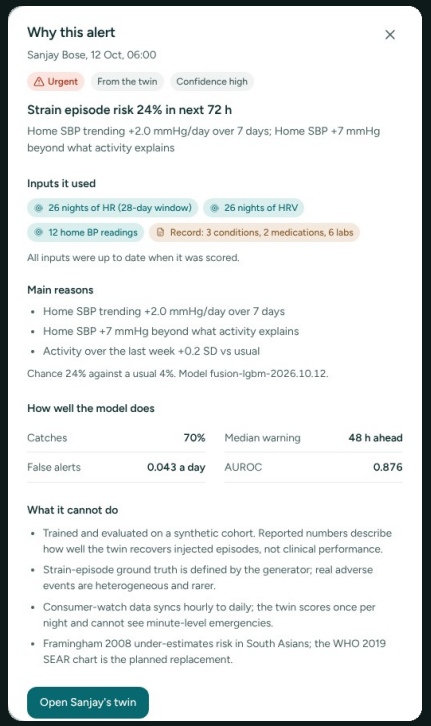
- **Today:** 10 urgent, 26 check-in, 204 steady, sorted by 72 h risk
- **Every row:** reason, sparklines, data freshness

- **Twin page:** three-sentence summary, 24% with its likely range
- **What the twin used** and a suggested next step: acknowledge, call, escalate

Doctor console: try changes on her body, keep alerts on budget

Simulate uses Meera's own slopes; every alert carries its reasons and inputs





0.037

false alerts a patient-day this week

0.045

budget the clinic sets

14 days

replayed before a threshold change

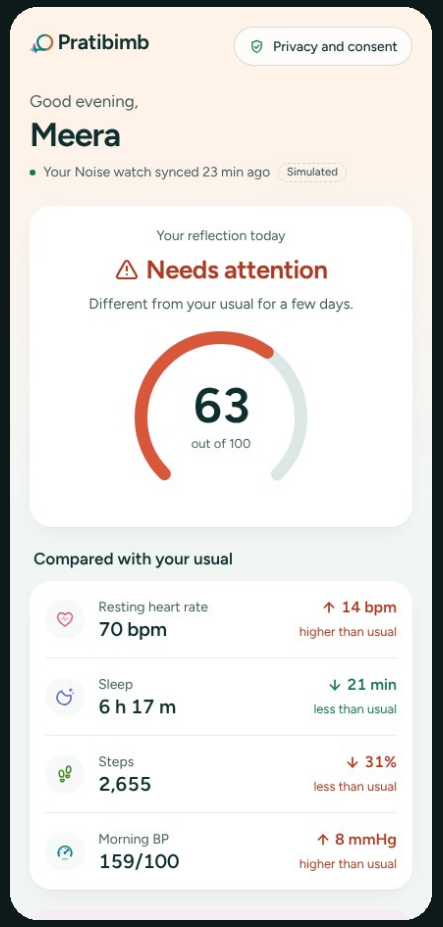
- **Simulate:** walking, sleep, medicine, salt, alcohol
- **72 h risk holds** once strain is under way; ordinary week falls 27%

- **Why this alert:** reasons, inputs, confidence
- **Model stats** beside every alert: 70% caught, AUROC 0.876

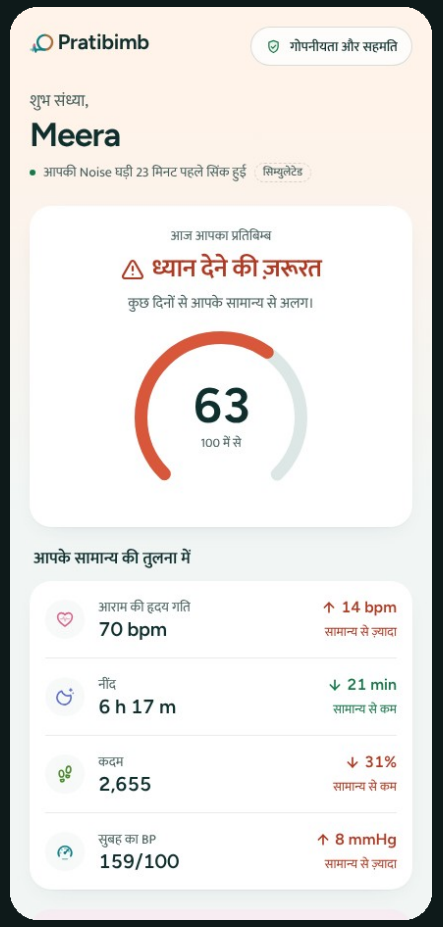
- **Thresholds:** preview on 14 days of history
- **Audit log:** append-only, with the consent basis

Patient view: plain words, her own levers, a safe SOS

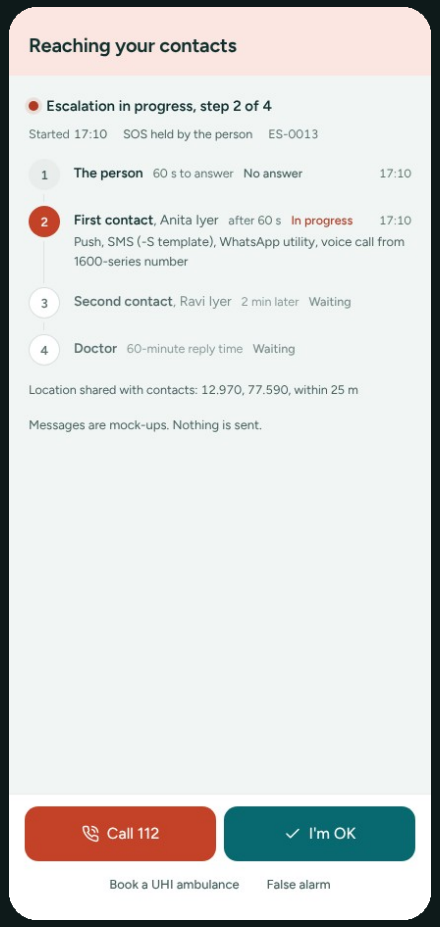
Live demo: pratibimb-pi.vercel.app/me/P-0007



Status vs her usual



Hindi, with read-aloud



SOS check-in ladder

- **One sentence** in English or Hindi; compared with her usual, not with others
- **What helps you:** her own strongest levers, learned from her calm days
- **Check-in:** fine, tired, dizzy, chest discomfort; a note to the doctor
- **Press-and-hold SOS:** patient first, then two contacts, then the doctor
- **One tap to 112** at every step; the server never dials it
- **New users:** "still learning your normal", never a guess

Built for India: devices, standards, consent, regulation

Details in docs/COMPLIANCE.md and docs/DEVICE_SYNC.md

Devices India owns

- **Budget watches** via Android Health Connect
- **Apple Watch** via HealthKit; Oura, Whoop; CSV upload
- **Freshness:** last sync shown on every card

Health standards

- **FHIR R4** with SNOMED CT and LOINC
- **ABDM WellnessRecord** bundle (NRCeS); masked ABHA
- **One-tap export** per patient from the Records tab

DPDP-ready consent

- **Consent per stream;** opt out of AI, keep records
- **Withdraw and delete;** 18+ gate; contacts consent ahead
- **Audit log:** consent basis of every action

Clinical safety

- **No AI prescribing:** patient text never changes doses (Telemedicine Guidelines 2020)
- **Doctor decides;** server never dials 112
- **Contacts reached** by SMS, WhatsApp, voice (mocked)

Regulatory path

- **CDSCO Medical Device Software** guidance (30 July 2026)
- **Likely Class B or C;** append-only audit log
- **Decision support** for a registered practitioner, not a diagnosis

Language and cost

- **English and Hindi** with read-aloud
- **Works on a Rs 2,000 watch** plus a home BP cuff
- **No new hardware:** reads the phone's health store

Outcomes: what is built and what it shows

Every item is in the public repository and the live demo

Delivered

Deliverable	Proof
Working prediction model	AUROC 0.876; 70% caught at budget; leakage-tested
Personal Physiology Engine	recovers hidden slopes, r 0.59 to 0.71
Doctor dashboard	triage, twin, simulate, alerts, thresholds, model card, audit
Patient app	English and Hindi, check-in, consented SOS ladder
Interoperability	FHIR R4 and ABDM WellnessRecord export
Live demo	pratibimb-pi.vercel.app, no sign-in
Open source	MIT; 25 tests; Docker; reproducible seed

A day in the clinic

12 October, 240 synthetic patients

10 need attention today

26 worth a check-in

204 steady: no action needed

0.041 false alerts a patient-day
inside the 0.045 budget; 36 open alerts, each with reasons and inputs

Honest limits and the road to a clinic

What the evidence does not show yet, and how we close each gap

Limits today

- **Synthetic labels:** real-world performance will be lower
- **Nightly, not minute by minute:** budget watches sync hourly to daily
- **No heart-attack detection;** acute events go to SOS and 112
- **Record adds ranking,** not timing; its job is reading the watch right
- **Framingham under-estimates** South Asian risk
- **Single splits vary** by up to 19 points; every headline is a 3-split mean

1	**Next 3 months**	ABDM sandbox integration; Android companion app on Health Connect
2	**3 to 6 months**	Silent-mode pilot with one hypertension clinic; validate on consented real data
3	**6 to 12 months**	WHO 2019 SEAR risk chart; CGM stream for diabetic patients; more Indian languages
4	**12 months on**	CDSCO Medical Device Software filing (Class B or C); clinic network roll-out

Team AMAZE, IIM Lucknow

Happiest Health Digital Twin Challenge 2026

Nikita Pandey

Team leader

PGPSM, IIM Lucknow

Suraj Aryan

Member

PGPSM, IIM Lucknow



Thank you

Pratibimb: a reflection of one body, read ahead of time.

Everything is public

Live demo	pratibimb-pi.vercel.app
Repository	github.com/pndynkt/Team-AMAZE_IIM-Lucknow
Algorithm	docs/ALGORITHM.md
Model card	docs/MODEL_CARD.md
Architecture	docs/architecture.pdf
Demo video	unlisted YouTube; link in the README

Research prototype on synthetic data. Not a medical device. In an emergency, call 112.